

Ex. 3.6

The file `m-mrk4608.txt` contains monthly simple returns of Merck stock from June 1946 to December 2008. The file has two columns denoting date and simple return. Transform the simple returns to log returns.

(a) Is there any evidence of serial correlations in the log returns? Use autocorrelations and 5% significance level to answer the question. If yes, remove the serial correlations.

(b) Is there any evidence of ARCH effects in the log returns? Use the residual series if there are serial correlations in part (a). Use Ljung–Box statistics for the squared returns (or residuals) with 6 and 12 lags of autocorrelations and 5% significance level to answer the question.

(c) Identify an ARCH model for the data and fit the identified model. Write down the fitted model.

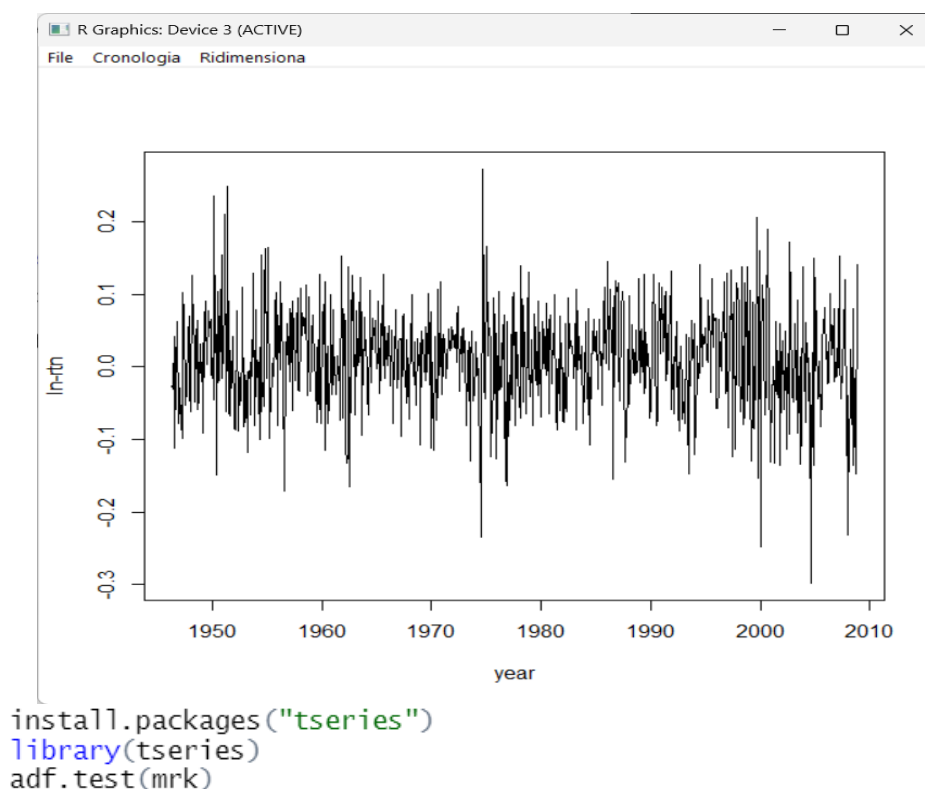
```
1 getwd()
2
3 da1 = read.table("m-mrk4608.txt", header = T)
4 head(da1)
```

First, the working directory is set and the dataset is imported into R:

```
> da1 = read.table("m-mrk4608.txt", header = T)
> head(da1)
      date      mrk
1 19460628 -0.025926
2 19460731 -0.030534
3 19460830  0.043307
4 19460930 -0.105660
5 19461031 -0.008475
6 19461130  0.064103

6 mrk= log(da1$mrk+1)
7 rtn = ts(mrk, frequency=12, start=c(1946,5))
8 windows()
9 plot(rtn, type= 'l', xlab= 'year', ylab= 'ln-rtn')
```

After importing the data, simple returns are converted into log returns. The resulting log return series is then converted into a time series object with monthly frequency and plotted in order to visually inspect its behavior:



From a first visual inspection, the log return series appears to fluctuate around a constant mean with no evident trend, suggesting stationarity. To formally assess stationarity, an Augmented Dickey–Fuller (ADF) test is performed: its result is consistent with expectations, as financial return series are typically stationary after logarithmic transformation.

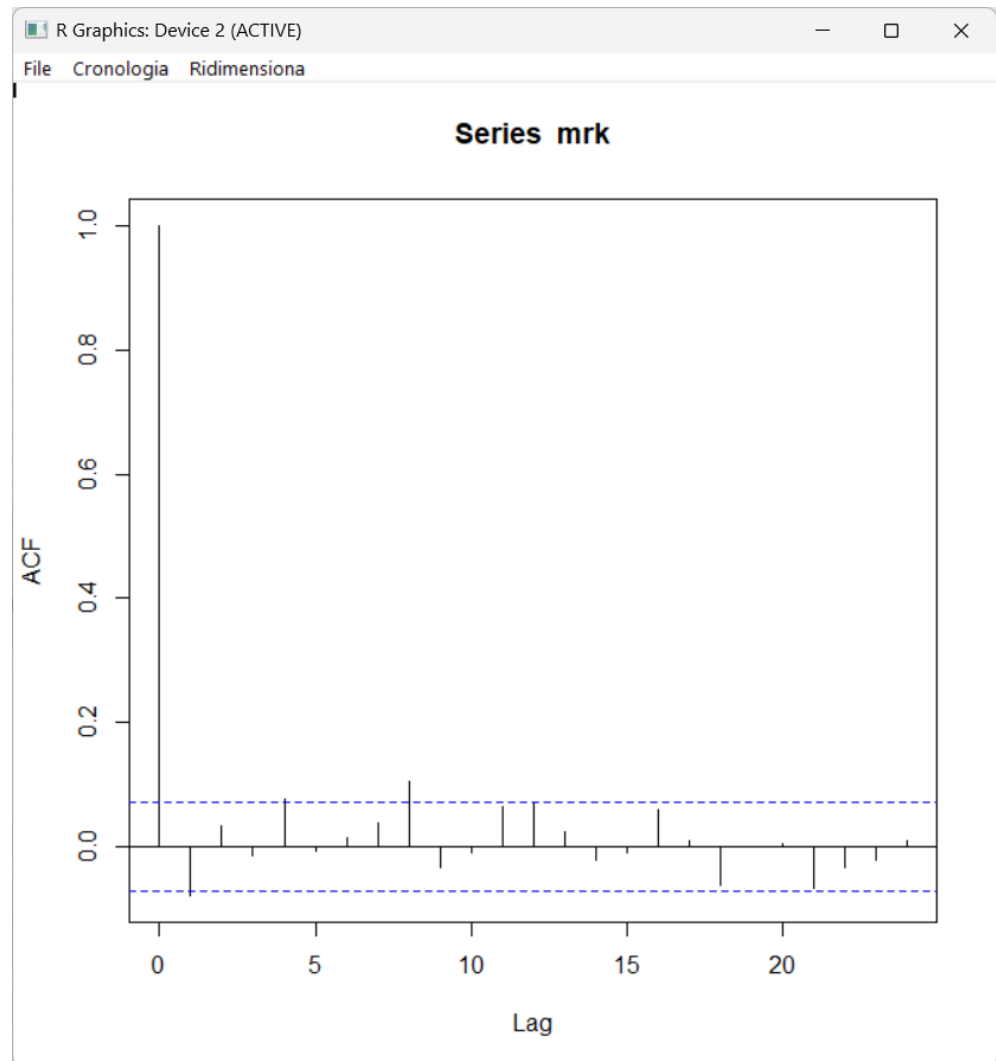
```

12 windows()
13 acf(mrk, lag=24)
14
15 Box.test(mrk, lag=12, type= 'Ljung')

```

To investigate the presence of serial correlation in the log return series, the sample autocorrelation function (ACF) is examined up to 24 lags. In addition, a Ljung–Box test is performed using 12 lags and a 5% significance level.

From the ACF plot, several autocorrelation coefficients appear to be statistically significant at specific lags (notably at lags 2, 4, and 9)



Box-Ljung test

```

data: mrk
X-squared = 27.236, df = 12, p-value = 0.007144

```

The p-value (<0.05) confirms what we said. These results indicate that the log return series exhibits statistically significant serial correlation. Consequently, it is appropriate to model and remove this dependence structure before proceeding with the analysis of conditional heteroskedasticity.

```

17 install.packages("forecast")
18 library(forecast)
19
20 m1 = auto.arima(mrk)
21 m1
22
23 windows()
24 res= residuals(m1)
25 acf(res, lag=24)
26 Box.test(res, lag=12, type= 'Ljung')

```

To eliminate serial correlation in the mean equation, a parsimonious ARMA model is fitted to the log return series using the `auto.arima()` function from the `forecast` package. The residuals from the selected ARMA model are then extracted and analyzed to verify whether serial correlation has been successfully removed.

```
> m1
Series: mrk
ARIMA(0,0,1) with non-zero mean
```

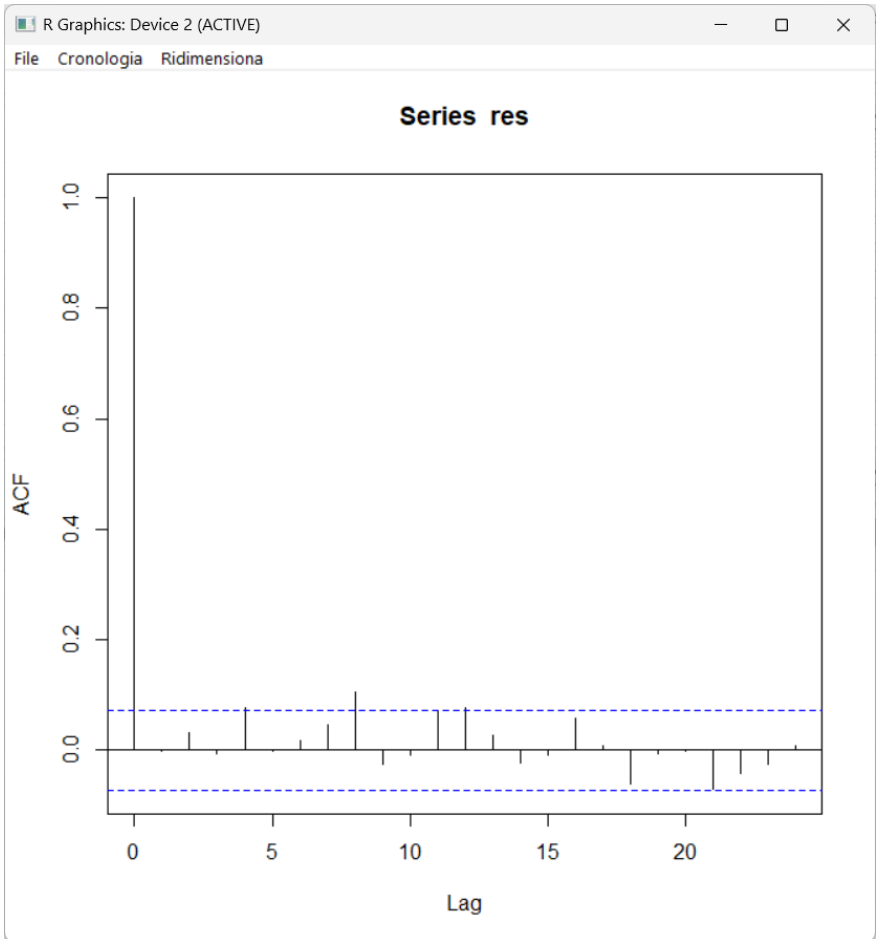
```
Coefficients:
      ma1      mean
-0.0744  0.0108
s.e.    0.0353  0.0024
```

```
sigma^2 = 0.004977:  log likelihood = 926.65
AIC=-1847.29  AICc=-1847.26  BIC=-1833.43
```

The `auto.arima()` procedure selects an ARMA(0,1) model with a non-zero mean for the log return series. However, as shown in the exhibit right below, the model doesn't fit the serie.

The ACF of the residuals does not display pronounced individual autocorrelations; however, the Ljung–Box test yields a p-value of 0.018, which is below the 5% significance level. Therefore, the null hypothesis of no serial correlation in the residuals is rejected.

This result indicates that the ARMA(0,1) specification is not sufficient to fully capture the linear dependence in the conditional mean of the log return series.



Box-Ljung test

```
data: res
X-squared = 24.335, df = 12, p-value = 0.01831
```

```
m_ar1 <- arima(mrk, order=c(2,0,2))
m_ar1
res_ar1= residuals(m_ar1)
Box.test(res_ar1, lag=12, type= 'Ljung')
```

Trying with an ARIMA(2,0,2), the ljung-box test ensures that the `m_ar1` residuals does not present autocorrelation. Therefore, the ARIMA(2,0,2) model adequately removes serial correlation from the conditional mean of the log return series and can be retained as the mean equation for subsequent analysis.

Box-Ljung test

```
data: res_ar1
X-squared = 19.254, df = 12, p-value = 0.08258
```

```

33 #testing arch effects
34 mrk_sq= ts(mrk^2, frequency= 12, start= c(1946,5))
35 mrk_sq
36
37 Box.test(mrk_sq, lag=12 ,type='Ljung')

```

Box-Ljung test

```

data: mrk_sq
X-squared = 29.207, df = 12, p-value = 0.003671

```

As expected and as the p-value confirms, the series presents ARCH effects.

```

Box.test(mrk_sq, lag=12 ,type='Ljung')

resARMA_sq =ts(res_ar1^2, frequency =12, start=c(1946,5))
Box.test(resARMA_sq, lag=12 ,type='Ljung')

```

The same is true for the residual².

Box-Ljung test

```

data: resARMA_sq
X-squared = 33.521, df = 12, p-value = 0.0008024

```

These results clearly indicate the presence of ARCH effects that are not explained by the mean equation. Consequently, a conditional variance model is required to adequately describe the volatility dynamics of the log return series.

```

46 #arch model building
47 at= mrk-mean (mrk)
48
49 Box.test(at^2 , lag=12, type='Ljung')
50
51 source("archtest.R")
52 archTest(at,12)
53
54 install.packages("fGarch")
55 library(fGarch)
56
57 windows()
58 par(mfcol=c(1,2))
59 acf(at^2, lag=24)
60 pacf(at^2, lag=24)
61
62 #fit an ARCH(3)
63 m2= garchFit(~1+ garch(3,0), data=res_ar1, trace=F)
64 m2
65
66 resi= residuals(m2, standardize= T)
67 Box.test(resi, lag=12, type='Ljung')
68 Box.test(resi^2, lag=12, type='Ljung')

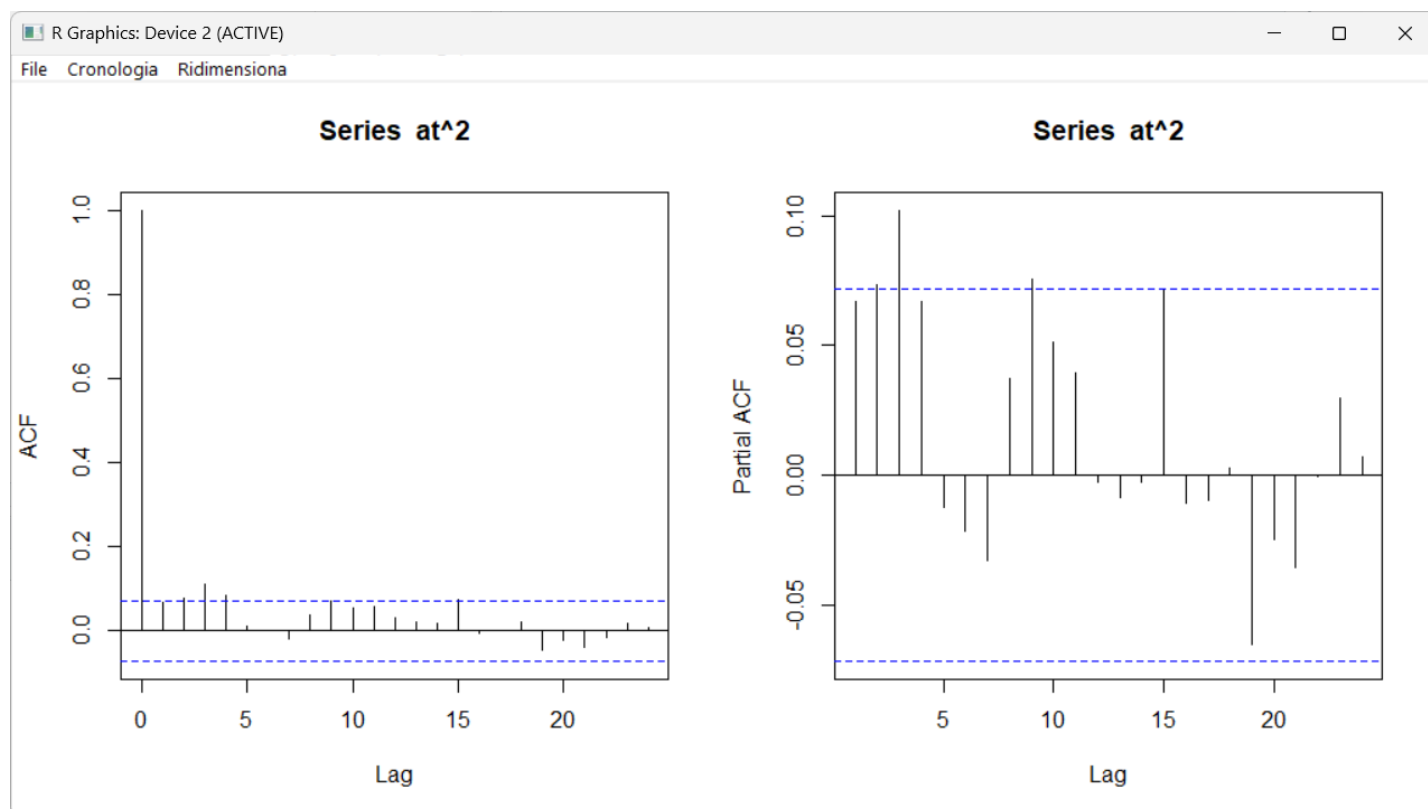
```

Steps to build the model:

- the first step is the specification of the mean equation.
- the second step is to test for arch effect (another time for completeness)
- the third step is the specification of an arch model
- the fourth step is to see if the model fit the series

Box-Ljung test

data: at^2
X-squared = 33.013, df = 12, p-value = 0.0009637



The ACF of the squared residuals shows significant autocorrelation at low lags, while the PACF exhibits a prominent spike at lag 3, suggesting an ARCH(3) specification for the conditional variance.

Based on this evidence, an ARCH(3) model is estimated using the `garchFit()` function from the `fGarch` package:

Mean and Variance Equation:

```
data ~ 1 + garch(3, 0)
<environment: 0x000001c11577cb78>
[data = res_ar1]
```

Conditional Distribution:

norm

Coefficient(s):

mu	omega	alpha1	alpha2	alpha3
0.00054798	0.00401122	0.02496372	0.06442203	0.09487936

Error Analysis:

	Estimate	Std. Error	t value	Pr(> t)
mu	0.0005480	0.0025240	0.217	0.8281
omega	0.0040112	0.0003262	12.298	<2e-16 ***
alpha1	0.0249637	0.0370574	0.674	0.5005
alpha2	0.0644220	0.0364818	1.766	0.0774 .
alpha3	0.0948794	0.0414192	2.291	0.0220 * 

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log Likelihood:

936.1979 normalized: 1.246602

To assess the adequacy of the fitted ARCH(3) model, standardized residuals are analyzed. In particular, Ljung–Box tests are applied to both the standardized residuals and their squares:

Box-Ljung test

data: resi

X-squared = 20.198, df = 12, p-value = 0.06343

> Box.test(resi^2, lag=12, type='Ljung')

Box-Ljung test

data: resi^2

X-squared = 11.69, df = 12, p-value = 0.4709

The p-values are both Greater than 0,05. This indicates that the ARCH(3) specification successfully captures the conditional heteroskedasticity present in the data.

Fitted model:

$$\varepsilon_t = \sigma_t z_t, \quad z_t \sim \text{i.i.d. } (0, 1)$$
$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \alpha_2 \varepsilon_{t-2}^2 + \alpha_3 \varepsilon_{t-3}^2$$

Ex. 3.8

The file `m-gmsp5008.txt` contains the dates and monthly simple returns of General Motors stock and the S&P 500 index from 1950 to 2008.

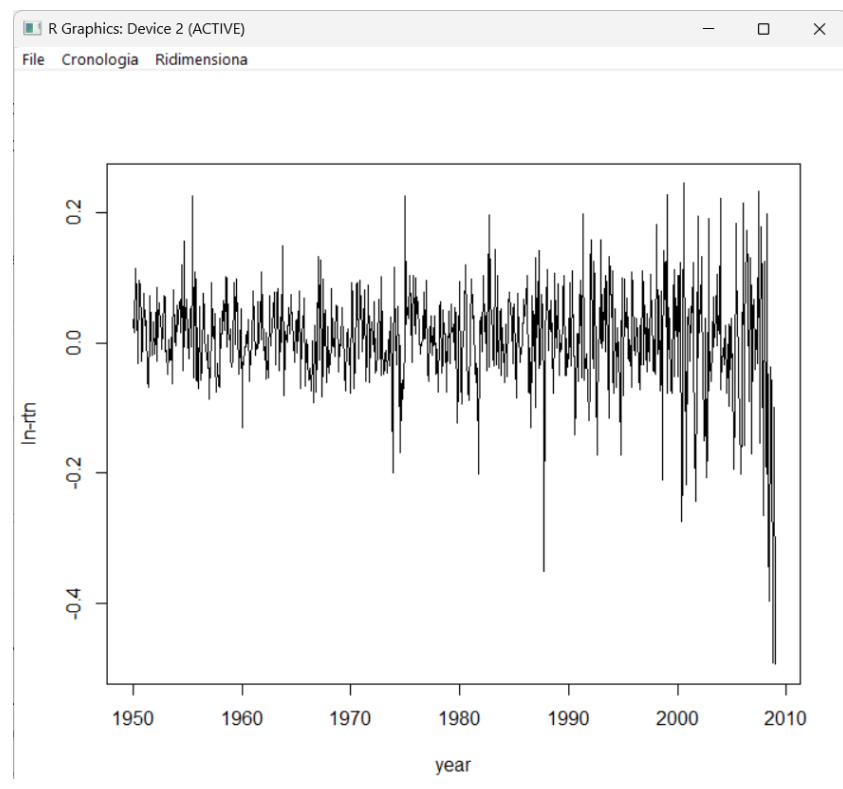
- Build a GARCH model with Gaussian innovations for the log returns of GM stock. Check the model and write down the fitted model.
- Build a GARCH-M model with Gaussian innovations for the log returns of GM stock. What is the fitted model?
- Build a GARCH model with Student- t distribution for the log returns of GM stock, including estimation of the degrees of freedom. Write down the fitted model. Let ν be the degrees of freedom of the Student- t distribution. Test the hypothesis $H_0: \nu = 6$ versus $H_a: \nu = 6$, using the 5% significance level.
- Build an EGARCH model for the log returns of GM stock. What is the fitted model?

```
75 #EX 3.8
76
77 da2 = read.table("m-gmsp5008.txt", header = T)
78 head(da2)
79
80 gm = log(da2$gm+1)
81 gm_ts = ts(gm, frequency = 12, start = c(1950,1))
82 windows()
83 plot(gm_ts, type= 'l', xlab= 'year', ylab= 'ln-rtn')
```

As before, we import the dataset as requested and transform the simple return into the log return.

A visual inspection of the series shows that the log returns fluctuate around a constant mean with no evident deterministic trend. Periods of high volatility are clearly visible, particularly toward the end of the sample during the 2008 financial crisis, suggesting volatility clustering

To formally assess stationarity, an Augmented Dickey-Fuller (ADF) test is performed: The test statistic yields a p-value equal to 0.01, which is below the 5% significance level. Therefore, the null hypothesis of a unit root is rejected, and the log return series can be considered stationary.



```
> adf.test(gm_ts)
```

Augmented Dickey-Fuller Test

```
data: gm_ts
Dickey-Fuller = -6.2666, Lag order = 8, p-value = 0.01
alternative hypothesis: stationary
```

```

86 #autocorrelation tests
87 Box.test(gm_ts, lag = 12, type = 'Ljung')
88 Box.test(gm_ts^2, lag = 12, type = 'Ljung')

```

Box-Ljung test

data: gm_ts
X-squared = 23.065, df = 12, p-value = 0.02718

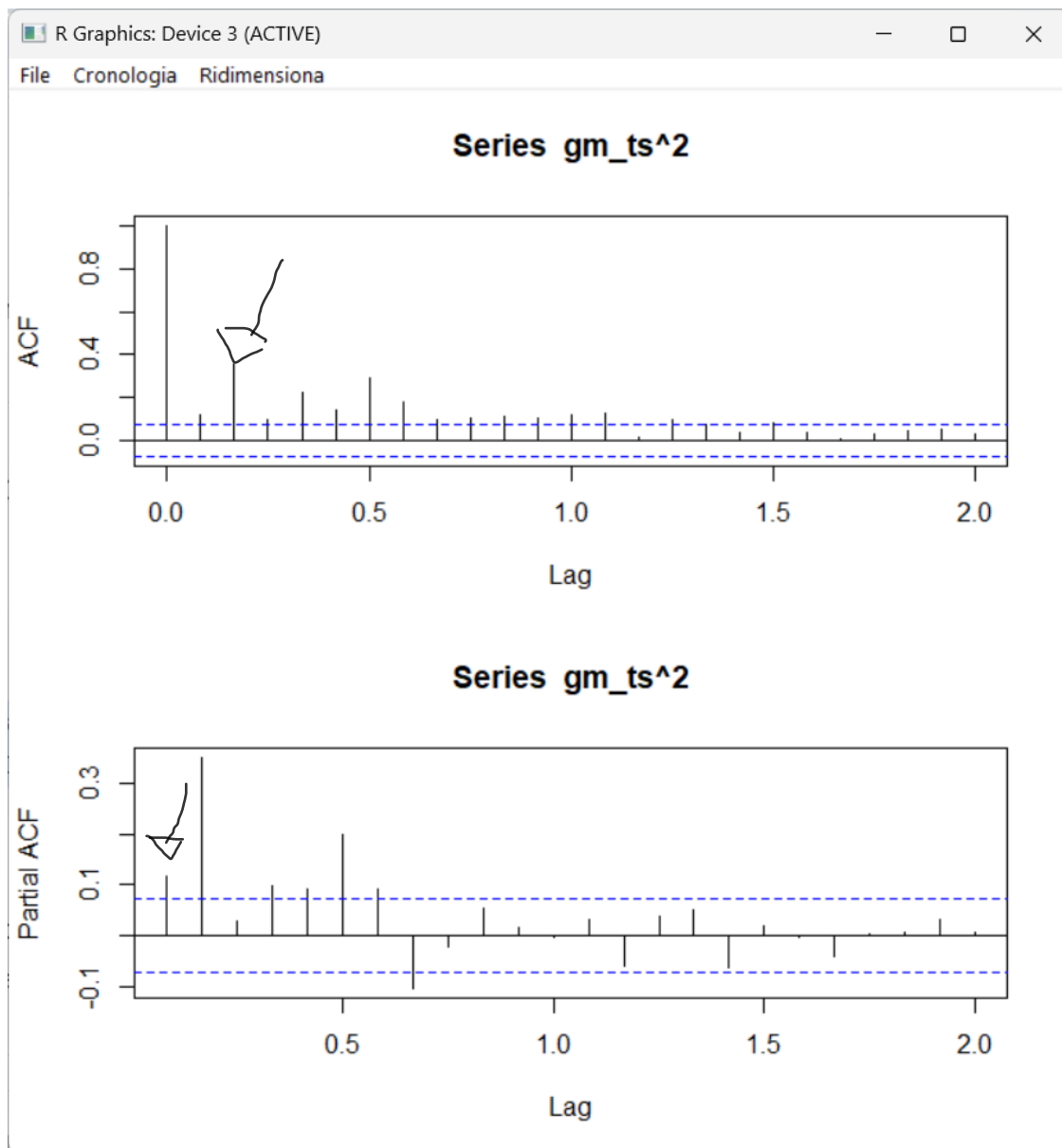
```
> Box.test(gm_ts^2, lag = 12, type = 'Ljung')
```

Box-Ljung test

data: gm_ts^2
X-squared = 285.24, df = 12, p-value < 2.2e-16

As shown by this test results, the series presents ARCH effect, which need to be modeled.

To guide the choice of an appropriate GARCH model, the autocorrelation function (ACF) and partial autocorrelation function (PACF) of the squared log returns are examined:



The ACF of the squared log returns displays significant autocorrelations at the first few lags, while the PACF shows a prominent spike at lag 1, followed by rapidly diminishing values. This pattern is characteristic of a low-order GARCH process.

Let's try with a GARCH(1,1).

```

95 m3= garchFit(~1+ garch(1,1), data= gm_ts, trace= F)
96 m3|

```

```

Coefficient(s):
      mu      omega      alpha1      beta1
0.00900750 0.00012171 0.11593859 0.87379548

```

```

Std. Errors:
based on Hessian

```

```

Error Analysis:
      Estimate Std. Error  t value Pr(>|t|)
mu      9.007e-03  2.403e-03   3.748 0.000178 ***
omega   1.217e-04  6.547e-05   1.859 0.063032 .
alpha1  1.159e-01  2.620e-02   4.424 9.67e-06 ***
beta1   8.738e-01  2.889e-02  30.244 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Let's check if the model fits the data:

```

98 res_m3= residuals(m3, standardize = TRUE)
99 Box.test(res_m3, lag = 12, type = 'Ljung')
100 Box.test(res_m3^2, lag = 12, type = 'Ljung')

```

Box-Ljung test

```

data: res_m3
X-squared = 15.934, df = 12, p-value = 0.1943

```

```
> Box.test(res_m3^2, lag = 12, type = 'Ljung')
```

Box-Ljung test

```

data: res_m3^2
X-squared = 12.833, df = 12, p-value = 0.3813

```

By looking at the p-values, we can assess that the GARCH(1,1) model adequately captures both the linear dependence in the mean and the conditional heteroskedasticity in the variance. The specification can be regarded as parsimonious and well fitted to the data.

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \sim \mathcal{N}(0, 1)$$

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

b)

We estimate a GARCH-M(1,1) model with Gaussian innovations for the log returns of GM stock. The term 'M' indicates that the conditional variance enters the mean equation as an additional regressor. The estimation is performed using the function `garchM()`, which implements the Gaussian maximum likelihood estimator.

```

102 source("garchM.R")
103 gm1 = log(da2$gm+1)
104 pm1 <- drop(as.matrix(gm_ts))
105
106 m6 = garchM(gm1, type = 1)
107 res_m6 =residuals(m6, standardize = TRUE)
108
109 Box.test(res_m6, lag = 12, type = 'Ljung')
110 Box.test(res_m6^2, lag = 12, type = 'Ljung')

```

The function `drop(as.matrix(x))` removes all time-series attributes from the data and converts it into a simple numeric vector, which is the required input format for the professor's GARCH-M estimation function.

```
> m6 = garchM(gm1, type = 1)
Maximized log-likelihood: 868.0837

Coefficient(s):
      Estimate   Std. Error   t value   Pr(>|t|)
mu      0.025157027 0.006259063 4.01930 5.8372e-05 ***
gamma -3.375284483 1.308327294 -2.57985 0.0098844 **
omega  0.000127048 0.000068297 1.86023 0.0628529 .
alpha  0.115938415 0.044816886 2.58694 0.0096834 **
beta   0.873795737 0.039706825 22.00619 < 2.22e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Box-Ljung test

```
data: res_m6
X-squared = 15.319, df = 12, p-value = 0.2244
```

```
> Box.test(res_m6^2, lag = 12, type = 'Ljung')
```

Box-Ljung test

```
data: res_m6^2
X-squared = 242.78, df = 12, p-value < 2.2e-16
```

The Ljung–Box test on standardized residuals indicates no significant autocorrelation ($p > 0.05$). However, the Ljung–Box test on squared residuals yields a very small p-value, showing that conditional heteroskedasticity has not been fully removed. Therefore, the GARCH-M(1,1) model is not adequate, as ARCH effects remain present in the standardized squared residuals.

c)

To address part (c), we estimate a GARCH model for the log returns of GM stock under the assumption of Student-t distributed innovations. The model is fitted via the `garchFit()` function with `cond.dist = "std"`.

```
112 m7= garchFit(~1+ garch(1,1), data= gm_ts, trace= F, cond.dist="std")
113 m7
114
115 t_stat = (9.418 - 6) / 2.968
116 t_stat
```

```
Coefficient(s):
      mu      omega      alpha1      beta1      shape
0.00865507 0.00012626 0.10419690 0.88243502 9.41766418
```

```
Std. Errors:
based on Hessian
```

```
Error Analysis:
      Estimate   Std. Error   t value   Pr(>|t|)
mu      8.655e-03 2.365e-03 3.659 0.000253 ***
omega  1.263e-04 8.067e-05 1.565 0.117530
alpha1 1.042e-01 2.820e-02 3.694 0.000220 ***
beta1  8.824e-01 3.445e-02 25.612 < 2e-16 ***
shape  9.418e+00 2.968e+00 3.173 0.001506 **
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Log Likelihood:
878.5255      normalized: 1.240855
```

The estimated degrees of freedom parameter is 9.418 with a standard error of 2.968. We compute the statistic shown in the line 115.

This is equal to 1,152, which is below the 5% critical value of 1.96. Therefore, we do not reject H_0 indicating that the estimated tail thickness is statistically consistent with $v = 6$.

d) To address part (d), we estimate an EGARCH(1,1) model for the log returns of GM stock using the function `Egarch()`. The model assumes normally distributed innovations and includes a mean term. The estimation procedure yields the parameter estimates reported below, together with their standard errors and associated test statistics.

```
118 source ("Egarch.R")
119 m8 = Egarch(gm1, include.mean = TRUE)
120 m8

> m8 = Egarch(gm1, include.mean = TRUE)
Maximized log-likelihood: 874.6753

Coefficient(s):
      Estimate Std. Error t value Pr(>|t|)
mu      0.00805545 0.00243980  3.30168 0.00096108 ***
alpha0 -0.21371281 0.07659448 -2.79019 0.00526779 **
alpha1  0.18689424 0.03864501  4.83618 1.3236e-06 ***
gamma  -0.12949746 0.11215849 -1.15459 0.24825696
beta    0.98657772 0.01107334 89.09488 < 2.22e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
122 res_m8 = residuals(m8, standardize = TRUE)
123 Box.test(res_m8, lag = 12, type = 'Ljung')
124 Box.test(res_m8^2, lag = 12, type = 'Ljung')
125 |
```

Box-Ljung test

```
data: res_m8
X-squared = 23.065, df = 12, p-value = 0.02718
```

```
> Box.test(res_m8^2, lag = 12, type = 'Ljung')
```

Box-Ljung test

```
data: res_m8^2
X-squared = 294.41, df = 12, p-value < 2.2e-16
```

The Ljung–Box test applied to the standardized residuals yields a p-value of 0.027, indicating residual autocorrelation at the 5% significance level. Furthermore, the Ljung–Box test on the squared standardized residuals produces a p-value $< 2.2e-16$, providing strong evidence of remaining ARCH effects. Taken together, these diagnostics show that the EGARCH(1,1) specification does not fully capture the serial dependence and volatility dynamics in the data, and therefore the model does not provide an adequate fit.

Ex. 3.9

Consider the monthly log returns of GM stock in `m-gmsp5008.txt`. Build an adequate TGARCH model for the series. Write down the fitted model and test for the significance of the leverage effect.

To address Exercise 3.9, a TGARCH(1,1) model was estimated for the monthly log-returns of GM stock using the `Tgarch11()` function. The estimation procedure converged and produced a set of parameter estimates for the conditional mean, the variance intercept, the ARCH component, the leverage term, and the GARCH component. However, the numerical Hessian matrix obtained at the optimum is nearly singular, which prevents the computation of reliable standard errors, t-statistics, and p-values. As a consequence, formal statistical inference on the individual coefficients—most notably on the leverage parameter—cannot be conducted.

Despite this limitation, the point estimates indicate that the leverage term is small, suggesting that any asymmetry in the volatility response to negative shocks is likely to be weak. Nonetheless, due to the singularity of the Hessian, the statistical significance of the leverage effect cannot be formally established.

```
127 #EX 3.9
128 |
129 source("Tgarch11.R")
130
131 m9 = Tgarch11(gm1,cond.dist = "norm", include.mean = TRUE)
132
133 res_m9 = residuals(m9)
134 Box.test(res_m9, lag = 12, type = 'Ljung')
135 Box.test(res_m9^2, lag = 12, type = 'Ljung')
```

```
> m9 = Tgarch11(gm1,cond.dist = "norm", include.mean = TRUE)
Log likelihood at MLEs:
[1] 866.8466
Conditional distribution used:
norm

Coefficient(s):
      Estimate Std. Error t value Pr(>|t|)
mu    0.008205142 0.002380023   3.4475 0.00056579 ***
omega 0.000393788      NaN      NaN      NaN
alpha 0.101015831      NaN      NaN      NaN
gam1   0.020951694 0.040921497   0.5120 0.60865292
beta   0.810754208      NaN      NaN      NaN
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Box-Ljung test

```
data: res_m9
X-squared = 23.065, df = 12, p-value = 0.02718
```

```
> Box.test(res_m9^2, lag = 12, type = 'Ljung')
```

Box-Ljung test

```
data: res_m9^2
X-squared = 294.53, df = 12, p-value < 2.2e-16
```

The test on the residuals yields a p-value of 0.027, which is below the 5% significance threshold, suggesting that the conditional mean equation is not fully capturing the linear dependence in the data. Moreover, the Ljung–Box test applied to the squared residuals returns an extremely small p-value ($< 2.2e-16$), providing strong evidence of remaining ARCH effects. This implies that the conditional variance dynamics specified by the TGARCH(1,1) structure are insufficient to account for the persistence and clustering in volatility.

These results closely mirror those obtained in Exercise 3.8 for the GARCH and EGARCH specifications. This similarity arises because monthly financial returns typically exhibit weak signal-to-noise ratios and persistent volatility clustering, which can be difficult to fully capture with low-order GARCH-type models. Consequently, alternative specifications—such as higher-order models, Student-t innovations, or models tailored to long-memory effects—may be required to achieve an adequate fit.

5. Suppose that ϵ_t is white noise with mean 0 and variance 1, that $a_t = \epsilon_t \sqrt{7 + a_{t-1}^2/2}$, and that $Y_t = 2 + 0.67Y_{t-1} + a_t$.

- (a) What is the mean of Y_t ?
- (b) What is the ACF of Y_t ?
- (c) What is the ACF of a_t ?
- (d) What is the ACF of a_t^2 ?

a) The process for Y_t is an AR(1) model with intercept 2 and autoregressive coefficient 0.67. Since $E[a_t]=0$, the unconditional mean is:

$$\mu_Y = \frac{2}{1 - 0.67} = \frac{2}{0.33} = 6.0606.$$

b) The autocorrelation function is:

$$\rho_k = (0.67)^k, \quad k = 1, 2, \dots$$

c) Since ϵ_t is Independent across time and Independent of a_{t-1} , it follows that:

$$\text{Cov}(a_t, a_{t-k}) = 0 \quad \text{for all } k \geq 1.$$

and the ACF

$$\rho_k(a_t) = 0 \quad \text{for all } k \geq 1.$$

d) a_t^2 depends directly on a_{t-1}^2 and a_t^2 is serially correlated. Thus:

$$\rho_k(a_t^2) \neq 0 \quad \text{for } k \geq 1.$$

6. Let Y_t be a stock's return in time period t and let X_t be the inflation rate during this time period. Assume the model

$$Y_t = \beta_0 + \beta_1 X_t + \delta \sigma_t + a_t, \quad (14.30)$$

where

$$a_t = \epsilon_t \sqrt{1 + 0.5a_{t-1}^2}. \quad (14.31)$$

Here the ϵ_t are independent $N(0, 1)$ random variables. Model (14.30)–(14.31) is called a *GARCH-in-mean* model or a GARCH-M model.

Assume that $\beta_0 = 0.06$, $\beta_1 = 0.35$, and $\delta = 0.22$.

- What is $E(Y_t | X_t = 0.1 \text{ and } a_{t-1} = 0.6)$?
- What is $\text{Var}(Y_t | X_t = 0.1 \text{ and } a_{t-1} = 0.6)$?
- Is the conditional distribution of Y_t given X_t and a_{t-1} normal? Why or why not?
- Is the marginal distribution of Y_t normal? Why or why not?

a) The conditional mean of the model is:

$$E(Y_t | X_t, a_{t-1}) = \beta_0 + \beta_1 X_t + \delta \sigma_t,$$

We compute the conditional standard deviation:

$$\sigma_t = \sqrt{1 + 0.5a_{t-1}^2} = \sqrt{1 + 0.5(0.6)^2} = \sqrt{1 + 0.18} = \sqrt{1.18}.$$

So:

$$E(Y_t | X_t = 0.1, a_{t-1} = 0.6) = 0.06 + 0.35(0.1) + 0.22\sqrt{1.18}.$$

$$\text{b) } \text{Var}(Y_t | X_t, a_{t-1}) = \text{Var}(a_t | a_{t-1}) = \sigma_t^2 = 1 + 0.5(0.6)^2 = 1.18.$$

c) Yes it is normal since

$$a_t = \epsilon_t \sigma_t \quad \text{with } \epsilon_t \sim N(0, 1)$$

and σ_t is measurable with respect to past information, it follows that

$$a_t | a_{t-1} \sim N(0, \sigma_t^2)$$

Because the mean equation is linear and deterministic conditional on X_t and a_{t-1} , we obtain:

$$Y_t | (X_t, a_{t-1}) \sim N(\beta_0 + \beta_1 X_t + \delta \sigma_t, \sigma_t^2)$$

d) No, the marginal distribution of Y_t is not normal because the conditional variance is time-varying and random, since it depends on σ^2_{t-1} .

A mixture of normals with different variances is not itself normal.